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Embedded Systems 1905330

Project Title: Automatic Rain Sensing Wiper System using Arduino

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1. Introduction

Today's car wipers are manual systems that work on the principle of manual switching. In our project , we present an automatic wiper system that automatically switches ON when detecting rain and stops when the rain stops.

In this project, we build the embedded system using an Arduino Uno as the microcontroller and a Breadboard combined with a rain sensor and two servo motors.

Our system is programmed to sense rain through a rain sensor, this signal is processed by a microcontroller to make the necessary action. The rain sensor operates on the principle of utilizing water to finish its circuit, therefore when it rains on it, the circuit closes and gives a signal to

the microcontroller. The microcontroller then processes this data and controls the motor.

We use Arduino C++ programming with the Arduino IDE to implement our system. We create the logic using the Servo library, which enables accurate positioning according to analog input values. The figure below shows the structure of the Proposed Model.

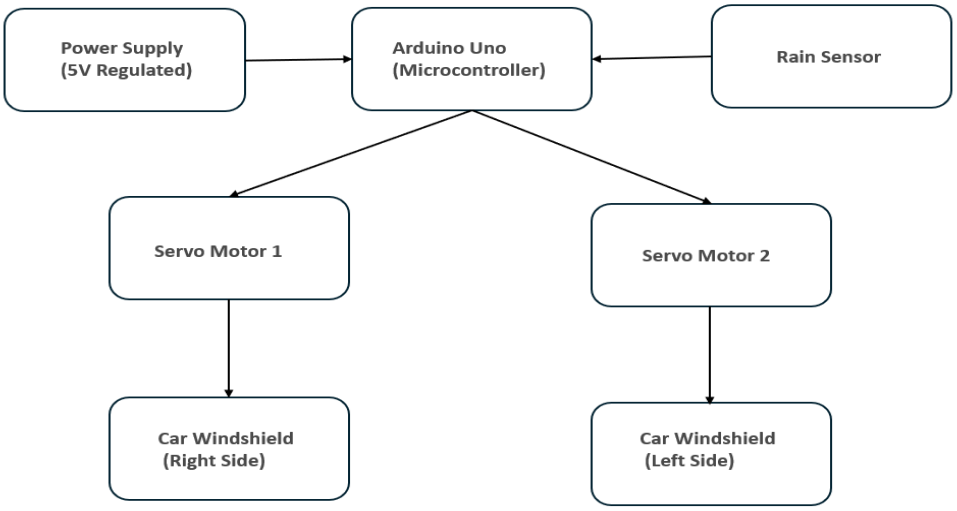

Figure 1: Automatic Rain-Sensing Wiper System - Block Diagram

2. Main flowchart of the system operation

The rain sensing car wiper system is an automated mechanism design to detect the intensity of rainfall by the implementation of basic conditional structures. It improves driver safety and comfort by responding in real time to changing weather conditions using basic conditional logic.

The flowchart below shows the operational logic for an automatic rain-sensing wiper system that operates two servos in response to rainfall intensity, as measured by an analog rain sensor. The operation starts by reading the analog input from the rain sensor, which returns an analog value from 0 to 1023 — with higher values representing dry and lower values representing heavy rain. These readings from the sensors are then compared to determine how the motor should work. After that, the system continually

loops back to read the sensor again so that it can Adjust in real time to the weather.

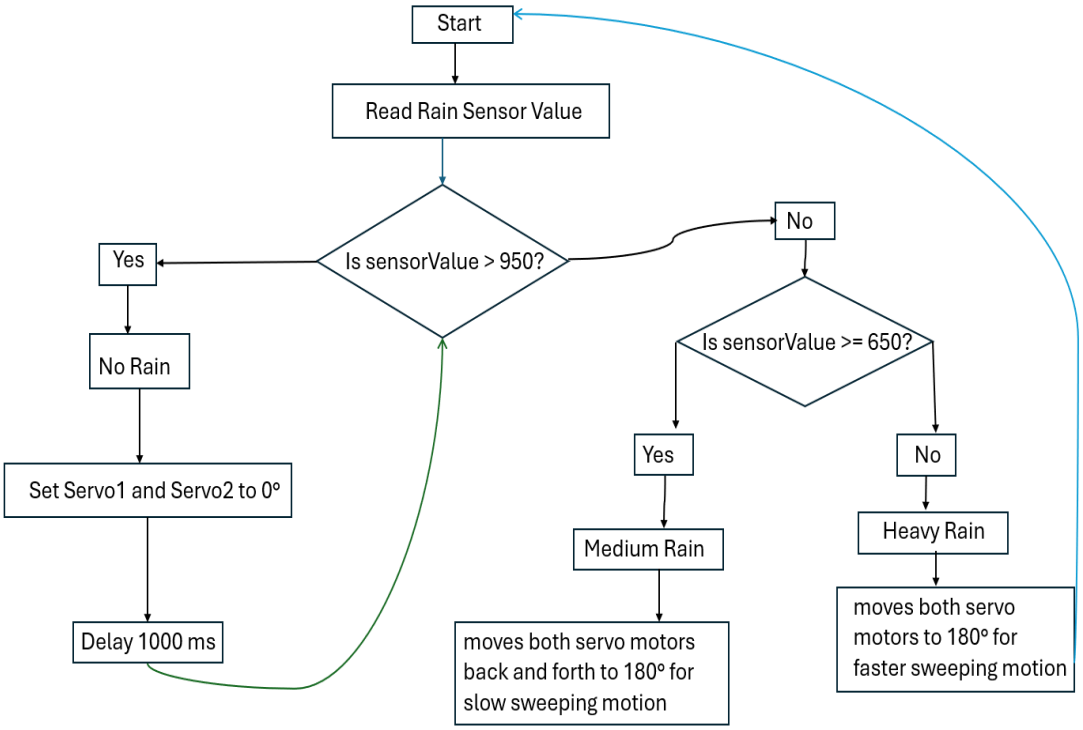

Figure 2: main flowchart of the Rain Sensing Car Wiper

The system operates as follows:

An analog rain sensor measures the amount of water present on its surface.

Based on the sensor reading, the microcontroller (in Figure 3) is constantly reading analog inputs from the rain sensor and comparing the values with predefined thresholds to determine if there is no rain, medium rain, or heavy rain. Depending on the outcome, it regulates the speed and engagement of the servo motors to modify the wiper movement accordingly.

Depending on the detected condition, the system:

- Turns off the servos (no rain).

- Move the servos slowly (medium rain).
- Moves the servos quickly (heavy rain).

Figure 3 : Arduino UNO

Main Flowchart Structure

At the start of the program, the following initialization steps are performed:

- Serial communication is initialized at 9600.
- Two Servo objects (wiperServo1 and wiperServo2) are created and attached to digital pins 9 and 10 respectively.
- Both servos are initialized to 0° position , indicating a static state.

This ensures that all hardware components are properly configured before entering the main loop.

Then, the core of the system runs in an infinite loop, continuously monitoring the environment and responding in real time. Here's a breakdown of the main loop logic:

Reading Rain Sensor Value

The system reads the analog value from the rain sensor using analogRead(A0).

The sensor returns a value between 0 and 1023 , where:

- High values (close to 1023) indicate dry conditions.
- Low values (close to 0) indicate wet conditions.

Displaying Sensor Reading

For debugging and monitoring, the current sensor value is printed to the Serial Monitor.

Determining Rain Level

The system uses conditional statements to classify the rain level into three categories:

- No Rain : Sensor value > 950
- Medium Rain : Sensor value between 650 and 950
- Heavy Rain : Sensor value < 650

Control Servos Based on Rain Level

Depending on the detected rain level, the system performs one of the following actions:

Case A: No Rain (Sensor Value > 950)

The servos are set to 0° , effectively turning them off.

A message “No Rain” is printed to the Serial Monitor.

The system waits for 1 second before taking the next reading to avoid unnecessary updates.

Case B: Medium Rain (Sensor Value Between 650 and 950

The function servoCtrl is called.

This causes both servos to sweep from 0° to 180° and back with a delay of 4 milliseconds between each degree increment, resulting in a slow sweeping motion .

A message “Medium Rain” is printed to the Serial Monitor.

Case C: Heavy Rain (Sensor Value < 650)

The function servoCtrl is called.

This causes both servos to sweep from 0° to 180° and back with a delay of 15 milliseconds, resulting in a faster sweeping motion.

3. Actuators and Transducers Used in the System

3.1 Actuators

3.1.1 Rain Detection Sensor (Sensor / Transducer)

Figure 4 : Rain Detection Sensor

Function:

Rainfall intensity sensor is utilized to check how much rain is falling. It is the primary input device of the system.

Working Principle:

The sensor works on the principle of measuring the variation in conductivity on the surface of the sensor. Conductivity between sensor contacts increases as raindrops build up. This is translated into an analog voltage, which is read by the Arduino as rainfall intensity.

Signal Conversion:

The sensor transduces a physical environmental quantity (rainfall) into an electrical quantity (analog voltage). The output signal is a function of the wetness level on the sensor surface detected.

Integration with System:

The sensor is also linked to the Arduino analog input pin A0(This pin gives an analog signal between the voltage supply from 5V to
0V). The sensor continuously transmits real-time analog data to the microcontroller so that it will operate according to changes
in the surroundings.[1]

3.2 Transducers

3.2.1 Servo Motors (Actuators)

Figure 5 : servo motor

Function:

The servo motors are the system actuators. They replicate the motion of windshield wipers by moving back and forth according to control signals from the Arduino.

Working Principle:

The Arduino sends Servo motors Pulse Width Modulated (PWM)

signals. The pulse width determines the angle of rotation of the servo motor shaft. This allows the servo position to be precisely controlled.[1]

Mechanical Output:

The servo motors are all programmed to sweep from 0° to 180°,
and vice versa, simulating the movement of real wiper blades.
This is in line with the rate of rainfall as detected by the sensor.
Rain Intensity Dependency:

- No Rain (ADC > 950):
  Thesensor is reading dryness. Therefore, servos are in their default po sition (0°), and there is no wiper activity.
- Medium Rain (650 < ADC ≤ 950):
  The sensor picks up moderate rain. The servo
  motors perform the sweep action at a reduced speed, providing a moderate wiping action.
- Heavy Rain (ADC < 650):
  The sensor monitors heavy rainfall. The servo
  motors run at higher speed, closing the gap between strokes and givin g quick wiping for visibility.

Conclusion

The world will one day move in self-driving cars is already evident in a series of functions that today's cars have begun to perform without human intervention. So the Automatic Rain Sensing Car Wiper System is a useful and technically advanced system which has to be fully realized significant safety, convenience as well cost effectiveness. This system strikes the balance of using an extremely simple combination of components –a rain detection sensor with servo motors and interfacing it into a virtually real time

weather tracking which can be adapted for windscreen wipers performed by Arduino Uno microcontroller opportunities. By automating the rain prep stage of a drive, in variable weather conditions when it can throw anyone overboard at any time.

Tech wise, the system is an excellent application putting embedded tech and automation easily within reach for everyday modern problems. The sensor has a transducer that processes the environmental conditions as electrical signals and send it to microcontroller for interpretation. The microcontroller (MCU) which plays the role as CPU measures rain intensity then gives appropriate control signals to servo motors. This signal then triggers a set of actuators to move mechanically, turning on the wipers at different speeds depending upon how much rain has been detected.

The development proves popular both for users and investors should also encourage mass adoption in an economic perspective carry some quite profound potential ramifications:

The Automatic Rain Sensing Car Wiper System cost lesser to afford has its design typical and efficient in all forms. The system is fabricated from inexpensive parts like an Arduino Uno, servo motors and a rain sensor with total assembly cost well under \$30, making it suitable for retrofitting to lower- and middle-of-the-range vehicles as either OEM or licensed add-on. Since it only functions when rain is detected, the system lessens unnecessary use of wipers which can result in reduced wear and longer life of wiper blades as well as related parts resulting to lower maintenance costs. Additionally, automated wiper control results in better visibility which increases road safety and could reduce accidents as well as insurance claims and health care costs. It aligns well with the emergence of smart and autonomous vehicles as well. The combination of this with its scalability and possible integration to IoT platforms in addition to weather API provides an interesting asset for some future vehicle systems, which could also bring benefits not just for manufacturers but the consumers as well transportation infrastructures.

References

[1]Rain Sensor : Circuit, Types, Working & Its Applications

[2] https://www.electrical4u.com/what-is-servo-motor/